DHCP

* Dynamic host configuration protocol.
* It automatically assigns IP addresses to device on a network.
* Reduce manual IP configuration
* ex = A laptop connects to Wi-Fi and receives IP address 192.168.1.10 automatically from the DHCP server
* Operates at the Appⁿ layer of the OSI model.
* Main purpose = Automatically assign IP settings (IP address, subnet mask, gateway, DNS)

CIDR

* Classless Inter Domain Routing
* It defines the network & host portion of an IP address using a prefix length.
* Makes IP allocation more efficient & supports route aggregation
* ex = 192.168.1.0/24 represents a network with 256 IP addresses.
* CIDR is an IP addressing/routing notation, not a protocol.
* Efficient IP address allocation and routing.

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* DHCP = Automatically gives IP addresses to devices.
* CIDR = Defines how IP addresses are organized and represented (e.g = /24, /16, /30).